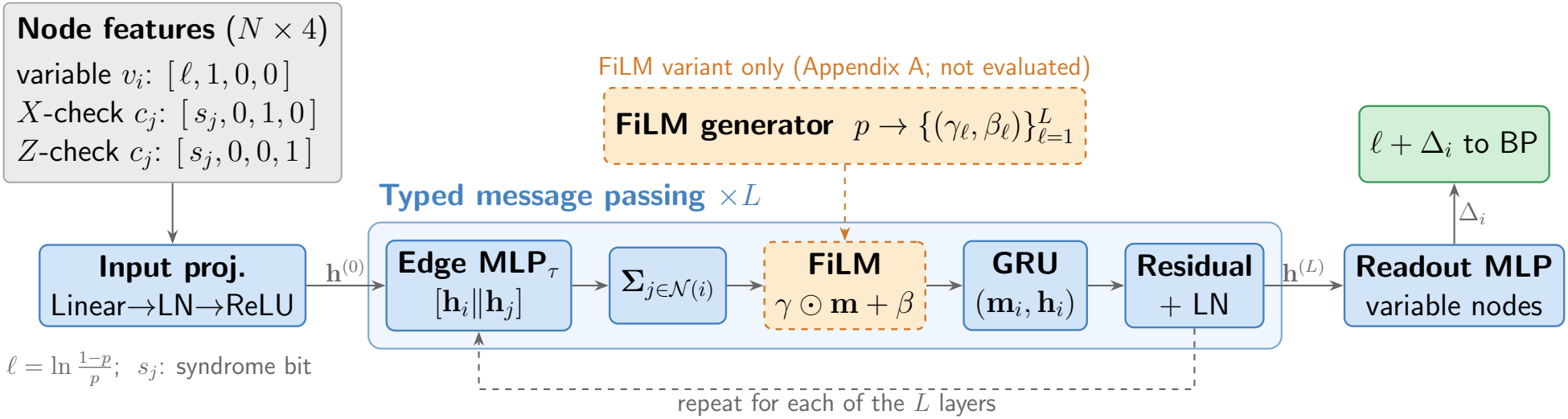

Node features $(N \times 4)$
variable v_i: $[\ell, 1, 0, 0]$
X-check c_j: $[s_j, 0, 1, 0]$
Z-check c_j: $[s_j, 0, 0, 1]$
$\ell = \ln \frac{1-p}{p}$; s_j: syndrome bit
FiLM variant only (Appendix A; not evaluated)
FiLM generator $p \to \{(\gamma_\ell, \beta_\ell)\}_{\ell=1}^{L}$
Typed message passing $\times L$
Input proj.
Linear→LN→ReLU
$\mathbf{h}^{(0)}$
Edge MLP$_\tau$
$[\mathbf{h}_i \| \mathbf{h}_j]$
$\Sigma_{j \in \mathcal{N}(i)}$
FiLM
$\gamma \odot \mathbf{m} + \beta$
GRU
$(\mathbf{m}_i, \mathbf{h}_i)$
Residual
+ LN
$\mathbf{h}^{(L)}$
Readout MLP
variable nodes
Δ_i
$\ell + \Delta_i$ to BP
repeat for each of the L layers